

Haokun Wang

Instructor, Radiation Oncology – Medical Physics & Engineering
UT Southwestern Medical Center

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Medical physics and engineering researcher with expertise spanning radiation oncology, electrical engineering, human-computer interaction (HCI), haptics, and extended reality (XR). My work combines engineering design, experimental methodology, computational analysis, and human-subject research to develop and evaluate novel technologies. I have extensive experience designing custom hardware and software systems, conducting quantitative and psychophysical studies, and translating interdisciplinary research into publications, demonstrations, and practical applications.

Current Appointment

2026 – Present **The University of Texas Southwestern Medical Center** – Dallas, TX, USA
Instructor, Radiation Oncology – Medical Physics & Engineering
Department of Radiation Oncology

Education

2019 – 2026 **The University of Texas at Dallas** – Richardson, TX, USA
PhD in Electrical Engineering
Advisor: Dr. Jin Ryong Kim

2016 – 2018 **University of Pittsburgh** – Pittsburgh, PA, USA
MS in Electrical Engineering
Advisor: Dr. Murat Akcakaya

2012 – 2016 **Tongji University** – Shanghai, China
BS in Electrical Engineering

Research Interests

Medical Physics & Engineering

Radiation oncology technology, medical imaging and instrumentation, computational modeling, experimental system development, human-centered medical technology

Human-Computer Interaction & XR

Haptics, thermal interfaces, wearable systems, virtual and augmented reality, multimodal interaction, psychophysics

Engineering

Custom hardware and PCB design, embedded systems, signal processing, electromagnetic simulation, thermal systems, mechanical prototyping, experimental instrumentation

Awards

- 2025 **Best Demo**, IEEE WHC 2025
- 2024 **Best Demo**, IEEE ISMAR 2024
- 2024 **Best Demo Honorable Mention**, IEEE Haptic Symposium 2024
- 2023 **Best Student Demo Award**, ACM SIGGRAPH Asia 2023
- 2023 **Best Paper Nominee**, IEEE VR 2023

Professional & Research Experience

- 2026 – Present **The University of Texas Southwestern Medical Center** – Dallas, TX, USA
Instructor, Radiation Oncology – Medical Physics & Engineering
Department of Radiation Oncology
- 2019 – 2026 **The University of Texas at Dallas** – Richardson, TX, USA
Graduate Research Assistant
- Designed and engineered multimodal haptic and wearable systems integrating thermal, tactile, and mechanical feedback for AR/VR applications.
 - Developed custom hardware and electronic systems, including PCBs, microcontroller-based platforms, motor drivers, wireless interfaces, thermal actuators, ultrasound haptic displays, gloves, and wearable interfaces.
 - Conducted controlled human-subject experiments and psychophysical studies to characterize perception, thresholds, sensory interactions, and user experience.
 - Applied quantitative analysis, statistical testing, ANOVA, regression, and psychophysical methods to evaluate experimental results.
 - Developed analytical and computational models of electromagnetic systems and experimentally validated model predictions using sensor-based measurements.
 - Led interdisciplinary research projects from concept development and engineering prototyping through experimental evaluation, publication, and public demonstration.
- 2016 – 2018 **University of Pittsburgh** – Pittsburgh, PA, USA
Graduate Research Assistant
- Developed sparse-recovery-based identification techniques using batch Orthogonal Matching Pursuit (OMP) for clutter-distribution identification in nonstationary environments.
 - Designed computational methods for signal analysis and evaluated algorithmic performance using experimental data.
- 2015 – 2016 **Tongji University** – Shanghai, China
Undergraduate Researcher
- Designed a closed-loop IGBT power-cycling test bench and developed an electro-thermal model to estimate semiconductor junction temperature.

Journal Publications

- J3. **Haokun Wang**, Yatharth Singhal, and Jin Ryong Kim. Let It Snow: Designing Snowfall Experience in VR. *Proc. ACM **IMWUT** 2024* (IF=4.5). DOI: <https://dl.acm.org/doi/abs/10.1145/3659587>
-  J2. Hyunki Son, **Haokun Wang**, Yatharth Singhal, and Jin Ryong Kim. Upper Body Thermal Referral and Tactile Masking for Localized Feedback. *IEEE **TVCG** 2023* (IF: 6.5). DOI: <https://ieeexplore.ieee.org/document/10049651>
- J1. Tianyu Chen, Carlos Caicedo-Narvaez, **Haokun Wang**, Mehdi Moallem, Babak Fahimi, and Morgan Kiani. Electromagnetic Compatibility Analysis of an Induction Motor Drive With Integrated Power Converter. *IEEE **TMAG** 2019* (IF: 1.9). DOI: <https://ieeexplore.ieee.org/document/8937059>

Conference Publications

- C7. **Haokun Wang**, Daniel Honrales, Hyunjae Gil, and Jin Ryong Kim. Thermal Masking Across the Human Body: Patterns, Pathways, and Perceptual Boundaries. *Proc. ACM **CHI** 2026* (Acceptance rate: 25.3%). DOI: <https://dl.acm.org/doi/abs/10.1145/3772318.3791601>
- C6. **Haokun Wang**, Yatharth Singhal, Hyunjae Gil, and Jin Ryong Kim. Fiery Hands: Designing Thermal Glove through Thermal and Tactile Integration for Virtual Object Manipulation. *Proc. ACM **UIST** 2024* (Acceptance rate: 24%). DOI: <https://dl.acm.org/doi/abs/10.1145/3654777.3676457>
- C5. Yatharth Singhal, Daniel Honrales, **Haokun Wang**, and Jin Ryong Kim. Thermal In Motion: Designing Thermal Flow Illusions with Tactile and Thermal Interaction. *Proc. ACM **UIST** 2024* (Acceptance rate: 24%). DOI: <https://dl.acm.org/doi/abs/10.1145/3654777.3676460>
- C4. **Haokun Wang**, Hyunjae Gil, and Jin Ryong Kim. Thermal Masking: When the Illusion Takes Over the Real. *Proc. ACM **CHI** 2024* (Acceptance rate: 26.4%). DOI: <https://dl.acm.org/doi/full/10.1145/3613904.3641941>
- C3. **Haokun Wang**, Yatharth Singhal, and Jin Ryong Kim. Fabric thermal display using ultrasonic waves. *Proc. IEEE **ISMAR** 2023* (Acceptance rate: 19.6%). DOI: <https://ieeexplore.ieee.org/document/10316365>
- C2. Yatharth Singhal, **Haokun Wang**, Hyunjae Gil, and Jin Ryong Kim. Mid-Air Thermo-Tactile Feedback using Ultrasound Haptic Display. *Proc. ACM **VRST** 2021* (Acceptance rate: 26.2%). DOI: <https://doi.org/10.1145/3489849.3489889>
- C1. Yijian Xiang; Malia Kelsey; **Haokun Wang**; Satyabrata Sen; Murat Akcakaya; Arye Nehorai. A Comparison of cognitive approaches for clutter-distribution identification in nonstationary environments. *Proc. IEEE **RadarConf18*** (Acceptance rate: 57.1%). DOI: <https://ieeexplore.ieee.org/document/8378604>

Conference Demonstrations

- D19. Haokun Wang, Ashish Pratap, Hyunjae Gil, Jun Sik Shin, Tae-Heon Yang, and Jin Ryong Kim. Demonstrating XRScissor: Magnetorheological Haptic Interface for Realistic Cutting in XR. IEEE *Haptic Symposium 2026*.
- D18. Haokun Wang, Yatharth Singhal, Hyunjae Gil, and Jin Ryong Kim. Demonstrating Fiery Hands: A Thermal-Tactile Thermal Glove using Thermal Masking. IEEE *Haptic Symposium 2026*.
- D17. Yatharth Singhal, Abbas Khawaja, Adam McCutcheon, Haokun Wang, Daniel Honrales, Anhaar Wasi, and Jin Ryong Kim. Demonstrating Moisture Transfer: A Perceptual Wetness Illusion Through Thermal and Wet Integration. IEEE *Haptic Symposium 2026*.
-  D16. Haokun Wang, Yatharth Singhal, Hyunjae Gil, and Jin Ryong Kim. Demonstration of Thermal Masking: When the Illusion Takes Over the Real. IEEE *WHC 2025*.
- D15. Haokun Wang, Yatharth Singhal, Hyunjae Gil, and Jin Ryong Kim. Demonstrating Fiery Hands: A Thermal-Tactile Thermal Glove using Thermal Masking. IEEE *WHC 2025*.
- D14. Yatharth Singhal, Daniel Honrales, Haokun Wang, and Jin Ryong Kim. Demonstration of Thermal Flow Illusions with Tactile and Thermal Interaction. IEEE *WHC 2025*.
- D13. Jun Sik Shin, Hyunjae Gil, Haokun Wang, Ashish Pratap, Shreya Krishnan, Woo Joo Kim, Gangyun Koh, Tae-Heon Yang and Jin Ryong Kim. Demonstrating XRScissor: Magnetorheological Haptic Interface for Realistic Cutting in XR. IEEE *WHC 2025*.
- D12. Yatharth Singhal, Daniel Honrales, Haokun Wang, and Jin Ryong Kim. Demonstration of Thermal Flow Illusions with Tactile and Thermal Interaction. IEEE *BHI 2024*.
- D11. Haokun Wang, Yatharth Singhal, Hyunjae Gil, and Jin Ryong Kim. Demonstration of Fiery Hands: Thermal Gloves through Thermal and Tactile Integration. IEEE *BHI 2024*.
- D10. Yatharth Singhal, Haokun Wang, and Jin Ryong Kim. Demonstration of FIRE: Mid-Air Thermo-Tactile Display. IEEE *BHI 2024*.
- D9. Yatharth Singhal, Daniel Honrales, Haokun Wang, and Jin Ryong Kim. Demonstration of Thermal Flow Illusions with Tactile and Thermal Interaction. IEEE *ISMAR 2024*.
- D8. Haokun Wang, Yatharth Singhal, Hyunjae Gil, and Jin Ryong Kim. Demonstration of Fiery Hands: Thermal Gloves through Thermal and Tactile Integration. IEEE *ISMAR 2024*.
- D7. Yatharth Singhal, Haokun Wang, and Jin Ryong Kim. Demonstration of FIRE: Mid-Air Thermo-Tactile Display. IEEE *ISMAR 2024*.
- D6. Haokun Wang, Yatharth Singhal, Hyunjae Gil, and Jin Ryong Kim. Fiery Hands: Designing Thermal Glove through Thermal and Tactile Integration for Virtual Object Manipulation. *Proc. ACM UIST 2024 Adjunct*.
- D5. Yatharth Singhal, Daniel Honrales, Haokun Wang, and Jin Ryong Kim. Thermal In Motion: Designing Thermal Flow Illusions with Tactile and Thermal Interaction. *Proc. ACM UIST 2024 Adjunct*.

- D4. Yatharth Singhal, **Haokun Wang**, and Jin Ryong Kim. FIRE: Mid-Air Thermo-Tactile Display. ACM **SIGGRAPH 2024**.
- D3. Yatharth Singhal, **Haokun Wang**, and Jin Ryong Kim. Mid-Air Thermo-Tactile Feedback using Ultrasound Haptic Display. IEEE **Haptic Symposium 2024**.
- D2. **Haokun Wang**, Yatharth Singhal, Hyunki Son, and Jin Ryong Kim. Demonstrating Upper Body Thermal Referral and Tactile Masking for Localized Feedback. IEEE **Haptic Symposium 2024**.
- D1. Yatharth Singhal, **Haokun Wang**, and Jin Ryong Kim. FIRE: Mid-Air Thermo-Tactile Display. ACM **SIGGRAPH 2024**.

Under Peer Review

- U1. Jun Sik Shin, Hyunjae Gil, **Haokun Wang**, Ashish Pratap, Shreya Krishnan, Woo Joo Kim, Gangyun Koh, Tae-Heon Yang, and Jin Ryong Kim. XRScissor: Magnetorheological Haptic Interface for Realistic Cutting. *IEEE TVCG 2026* (IF: 6.5).

Research & Technical Skills

Quantitative Research & Experimental Methods

- Human-subject research and experimental study design
- Psychophysical methods and threshold detection
- Null hypothesis significance testing (NHST)
- ANOVA and regression analysis using Python and MATLAB

Programming & Computational Tools

Python, C, C#, MATLAB, Verilog, JavaScript
Applied machine learning/AI, PyTorch, Scikit-learn

Electrical & Embedded Systems

PCB design (Altium), microcontrollers (Arduino, STM32, Raspberry Pi), FPGA
Signal processing, motor control, wireless systems, sensor integration
Electromagnetic simulation (Ansys Maxwell, HFSS)

Mechanical & Thermal Engineering

3D printing, CAD (Autodesk Fusion 360, Ansys SpaceClaim)
Ansys Mechanical, Fluent, MATLAB Simulink
Peltier-based thermal systems, haptic actuators (ERM, LRA, piezoelectric)
Machining: lathe, mill, drill, welding

XR & Prototyping

Unity, Quest HMD, OptiTrack, Leap Motion
Interaction design, usability evaluation, Figma
React/Next.js, Adobe Creative Cloud

Professional Service

2023 – Present

Peer Reviewer

ACM CHI, ACM ISMAR, ACM VRST, ACM DIS, ACM TEI, IEEE WHC, IEEE EuroHaptics, IEEE Haptic Symposium, IEEE Transaction on Haptics, Journal of Thermal Stresses.

References

Supervisor

Dr. Yulong Yan

Computational Physics Director

Department of Radiation Oncology

UT Southwestern Medical Center, USA

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Ph.D. Advisor

Dr. Jin Ryong Kim

Assistant Professor

Department of Computer Science

The University of Texas at Dallas, USA

✉ jin.kim@utdallas.edu

M.S. Advisor

Dr. Murat Akcakaya

Associate Professor

Department of Electrical Engineering

University of Pittsburgh, USA

✉ akcakaya@pitt.edu

Collaborator

Dr. Hyejin Kim

Principal Researcher

ICT Materials and Components Research Laboratory

Electronics and Telecommunications Research Institute, South Korea

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Dr. Tae-Heon Yang

Associate Professor

Department of Mechanical Engineering

Konkuk University, South Korea

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